Almost any kind of data can be stored in a blockchain but in the case of cryptocurrencies, data related to the user and transaction data are mainly stored, which is also known as a distributed ledger. Cryptocurrencies like Bitcoin use a concept called 'proof of work', which is carried out usually by Bitcoin miners, in order to validate any transaction over the blockchain network.

Bitcoin miners use powerful computing machines (ASIC processors) to solve certain mathematical problems in a network. Such machines are like supercomputers, specially designed for processing data over a Bitcoin network. The efficiency of a mining machine is measured in terahashes and today, a speed of at least 12 terahash/sec is required for mining Bitcoins. The more powerful the miner's system is, the higher the chances of solving the math problems, and the miner who solves it first, gets the chance to create the next block and validate any transaction on the Bitcoin network.

However, the proof-of-work (PoW) concept requires a lot of computing power, and the ASIC machines use a lot of electrical power too. Therefore, some cryptocurrencies use proof-of-stake (PoS), where the creator for the next block is randomly chosen based on the wealth and age (i.e., the stake); but such cryptocurrencies are more vulnerable to fake stake attacks, because the attacker can cash an affected node by using very little or no stake.

Blockchain layers

A blockchain operates mainly through four layers – the user interface (UI) layer, the application layer, the consensus/validation layer and the data layer.

The UI layer is the topmost one, and it helps the user interact with the blockchain network. The application layer is the second one, and any app that uses a blockchain operates in this layer. Then comes the consensus/validation layer. The instructions sent by the apps are validated in

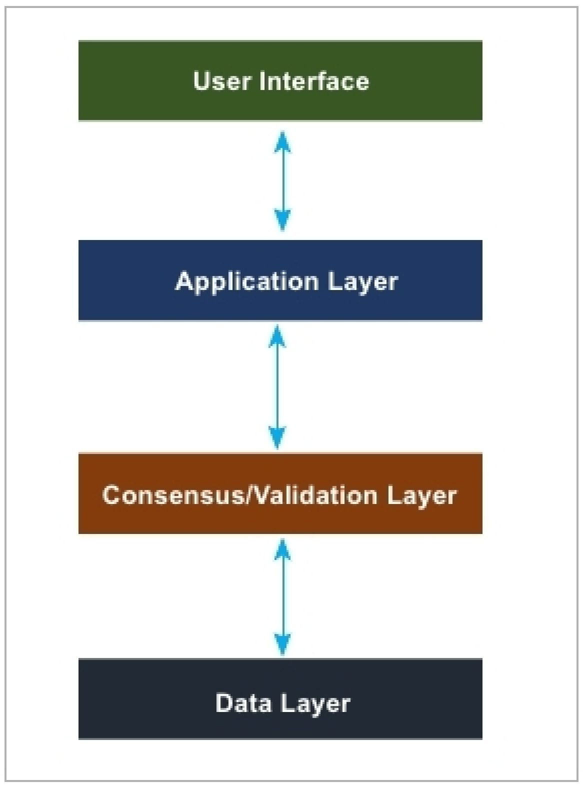

Figure 2: Blockchain layers

this layer by the algorithms of the blockchain network and in the case of cryptocurrencies, proof-of-work (PoW) and proof-of-stake (PoS) work in this layer. The final layer is the data layer, where the processing of data, like the storage and retrieval of data from the blocks, is done.

Using a blockchain during a pandemic

Blockchain can be very useful in many ways during a pandemic like COVID-19, while also being used for cryptocurrency, supply chain management (SCM) and infection tracking. As blockchain seems more reliable and secured, using a blockchain network for COVID-19 can help people manage their lives and also control the pandemic.

- **Cryptocurrencies:** Using cryptocurrencies based on blockchain during and after COVID-19 can help people make financial transactions that are reliable and flawless. As the coronavirus can also be transmited through currency notes and since social distancing is the primary preventive measure for COVID-19, using cryptocurrencies like Bitcoin can help control the pandemic. When compared with other digital payments like debit/credit cards, electronic fund transfers (EFT)/ wire transfers and online wallets, cryptocurrencies on blockchain networks are more reliable and less prone to hacking. Moreover, centralised online payment services could go down because of a technical problem or virus attack, but as a blockchain is always a decentralised network, cryptocurrencies like Bitcoin are less vulnerable to such problems. Maybe because of such benefits, the Supreme Court has lifted the ban on Bitcoin in India, which was imposed by the RBI in 2018.

  The value of one Bitcoin has remained above US$ 5000 (₹ 400,000) for the last one year and reached its highest value (US$ 12,927.40) on June 26, 2019. One can buy and trade Bitcoins in India through various trading companies like BuyUcoin and Zebpay. Bitcoin uses a Secure Hash Algorithm (SHA) for its blockchain network and, as the name suggests, it's a secure algorithm that prevents any kind of hacking attacks. Apart from Bitcoin, one can also trade other cryptocurrencies like Bitcoin Cash, Ethereum and Litecoin on such platforms. However, not all cryptocurrencies use proof-of-work (PoW). Currencies like Peercoin, Nxt and PotCoin use either proof-of-stake (PoS) or both. As PoS is not as reliable as PoW, Bitcoin is much more reliable when compared with other cryptocurrencies that use only proof-of-stake (PoS). Facebook is planning to launch its own digital currency called Libra, which will use blockchain for any transaction and is based on an open source model.
- **Supply chain management (SCM):** The delivery of essential items like food and medicines is important during any pandemic and it's SCM that makes this possible. SCM is the process of keeping track of goods and inventories in a way that could help the logistics team to deliver